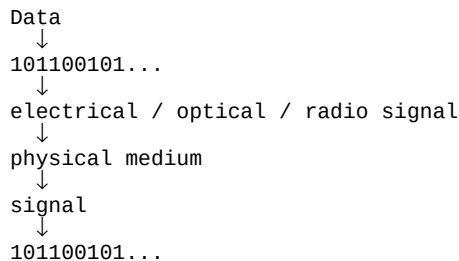


OSI Model – L1, L2 and L3

Principle of operation of the Physical, Data Link and Network layers

L1 – Physical Layer

Purpose: Transmit **bits (0 and 1)** over a physical medium. L1 does not care who the data belongs to; it deals with the physical transmission of the signal.



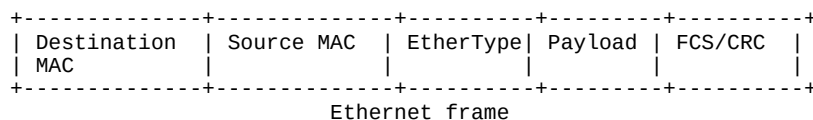
Examples: UTP cables, optical fiber, Wi-Fi radio, connectors, electrical signals, modulation, link speed and duplex.

L1 Devices

Device	Purpose
Repeater	Regenerates and repeats a physical signal.
Hub	Repeats the received signal to other ports; does not use MAC addresses.
Media converter	Converts between physical media, e.g. Ethernet and fiber.
PHY / transceiver	Converts between digital logic and the physical signal.
Modem	Modulates and demodulates signals.
Cables / passive components	UTP, fiber, patch panels and connectors.
Antennas	Transmit and receive radio signals.

L2 – Data Link Layer

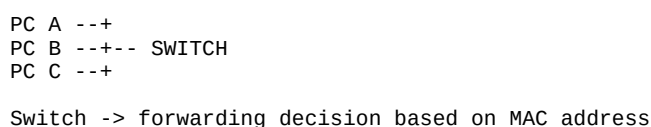
Purpose: Transfer data between **directly connected devices** and organize it into **frames**. L2 primarily works with **MAC addresses**.



L2 handles: MAC addresses, Ethernet frames, switching, VLAN (802.1Q), STP, FCS/CRC and broadcast domains.

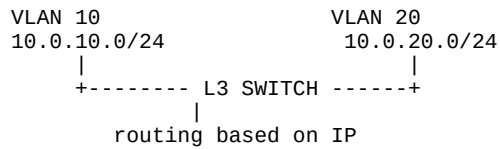
Switching at L2

A standard Ethernet switch is primarily an **L2 device**. It receives an Ethernet frame, checks the **destination MAC address**, and uses its MAC address table to decide which port should receive the frame.



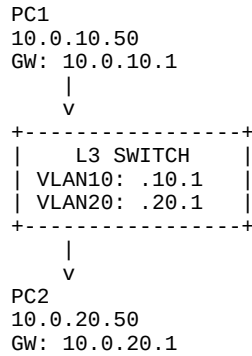
L3 – Network Layer

Purpose: Route IP packets between different networks. L3 works with **IP addresses** and a **routing table**.



L3 Switch

An L3 switch combines L2 switching and L3 routing. It can switch Ethernet frames using MAC addresses and route IP packets between different networks or VLANs.



When traffic moves between VLAN 10 and VLAN 20, the L3 switch uses its routing table, determines the destination network and forwards the packet through the appropriate L3 interface. A new Ethernet frame is created for the next hop.

Quick Comparison

Device	Primary layer	Makes decisions using
Hub	L1	physical signal
Switch	L2	MAC address
L3 switch	L2 + L3	MAC + IP / routing table
Router	L3	IP addresses / routing table

Remember: L1 = transmit bits. L2 = deliver frames using MAC addresses on the local network. L3 = deliver packets between IP networks.